

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech Branch: Computer Engineering /Computer Science & Engineering

Semester: VII

Subject Code & Name: BTCOE703A_Y20 Cloud Computing

Max Marks: 60

Date: 23/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is NOT a characteristic of cloud computing? a) On-demand self-service b) Rapid elasticity c) Localized processing d) Broad network access		1
2	VMware and Microsoft Hyper-V are examples of which technology in cloud computing? a) Cloud Interoperability b) Virtualization c) Cloud Architecture d) Cloud Scalability		1
3	Before moving applications to the cloud, it is essential to: a) Ignore software licensing b) Understand cloud cost models and software licenses c) Avoid checking service levels d) Skip cost-benefit analysis		1
4	Which cloud service model provides a complete application or software solution? a) IaaS b) PaaS c) SaaS d) CaaS		1
5	Which type of cloud deployment is owned and operated by a single organization exclusively? a) Public Cloud b) Hybrid Cloud c) Community Cloud d) Private Cloud		1
6	The capability to dynamically handle increased load by adding resources is known as: a) Cloud Interoperability b) Scalability c) Fault Tolerance d) Disaster Recovery		1
7	Disaster Recovery Planning in cloud computing primarily addresses: a) Security breaches only b) Cloud service downtimes and data losses c) Database design issues d) Application scalability challenges		1

8	Which service involves providing cloud users with complete business logic or business process? a) Database as a Service b) Storage as a Service c) Process as a Service d) Information as a Service		1
9	The Aneka platform primarily provides which of the following functionalities? a) Cloud programming and management tools b) Only virtual machine creation c) Disaster recovery solutions only d) Network infrastructure solutions exclusively		1
10	The layer responsible for managing resources in Aneka platform is called: a) Platform Abstraction Layer b) Application Services Layer c) Fabric Services Layer d) Foundation Services Layer		1
11	Amazon EC2 primarily falls under which category of cloud services? a) Communication services b) Compute services c) Storage services d) Additional services		1
12	Which cloud platform uses SQL Azure for cloud-based database management? a) Amazon Web Services b) Google AppEngine c) Microsoft Azure d) Aneka		1
Q. 2	Solve the following.		12
A)	Explain the characteristics and benefits of cloud computing. How does cloud computing differ from traditional computing paradigms?		6
B)	Describe virtualization and its characteristics. Discuss the taxonomy of virtualization techniques.		6
Q.3	Solve the following.		12
A)	Illustrate and describe the cloud computing reference model with suitable diagrams.		6
B)	Explain the importance of scalability and fault tolerance in cloud computing. How do these aspects enhance cloud reliability?		6
Q. 4	Solve Any Two of the following.		12
A)	Discuss the following cloud services with suitable examples: a) Storage as a Service b) Database as a Service		6
B)	What is Disaster Recovery in cloud computing? Explain the disaster recovery planning process and disaster management strategies.		6
C)	Define "Cloud Capacity Planning" and discuss how organizations can effectively scale their cloud infrastructure.		6
Q.5	Solve Any Two of the following.		12
A)	Describe the Aneka framework, mentioning its primary components and their roles.		6

B)	Explain the difference between Private, Public, and Hybrid cloud deployment modes in Aneka with suitable use cases.		6
C)	Discuss the Aneka Software Development Kit (SDK) and explain how Aneka facilitates cloud programming and management.		6
Q. 6	Solve Any Two of the following.		12
A)	Illustrate how cloud computing is advantageous in scientific applications, specifically focusing on healthcare and geo-sciences.		6
B)	Describe the architecture, core concepts, and cost model of Google AppEngine.		6
C)	Discuss Microsoft Azure's cloud platform, emphasizing Azure Core Concepts and SQL Azure services with suitable examples.		6
	*** End ***		